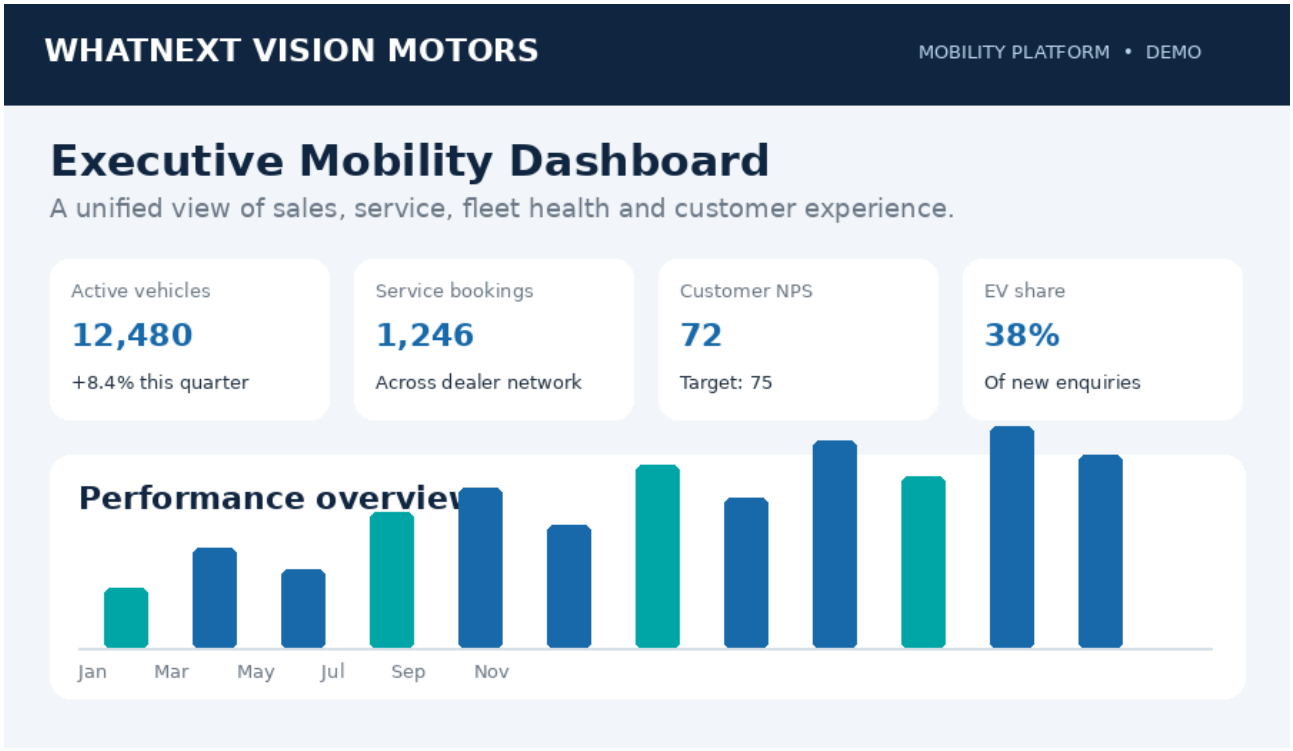
WHATNEXT VISION MOTORS

Shaping the Future of Mobility with Innovation and Excellence

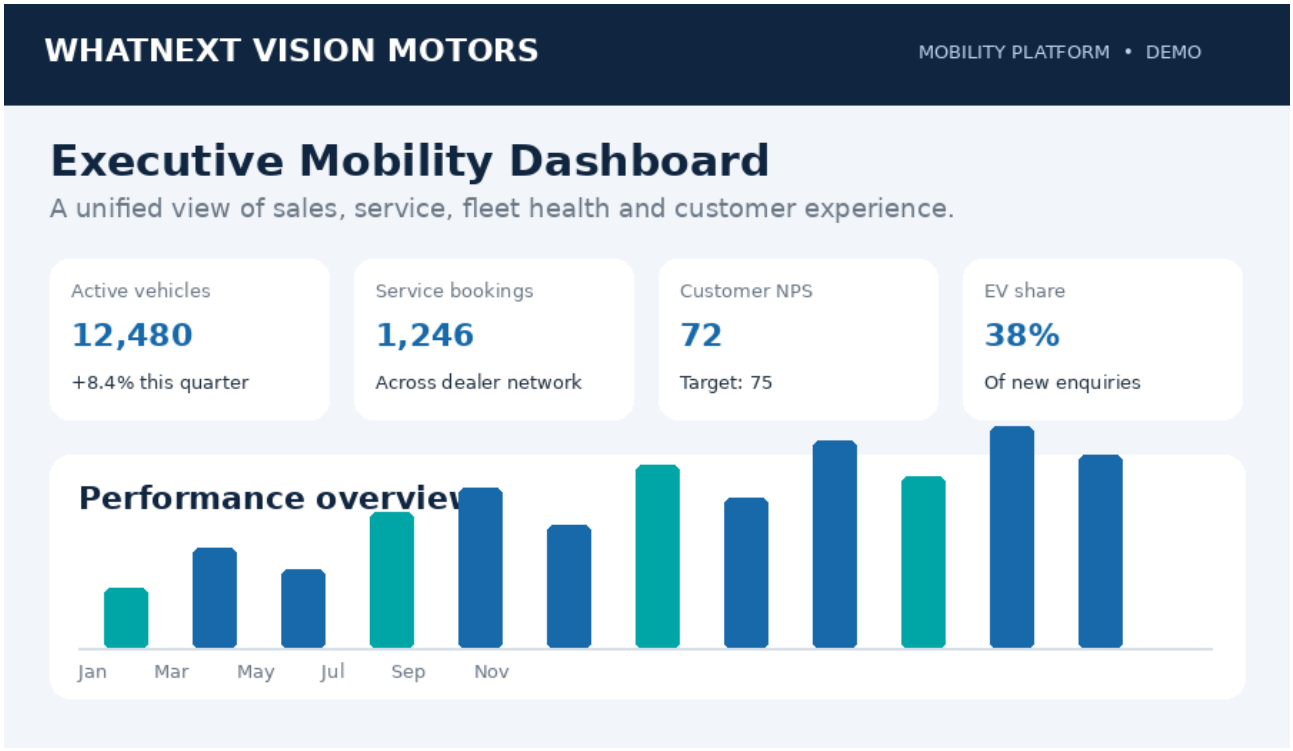
Project Proposal • Use Cases • Requirements • Delivery Milestones • UI Screenshot Concepts



Prepared as a planning document for a proposed digital mobility platform. Product names, metrics, screens and timelines are illustrative and should be validated with stakeholders.

1. Executive Summary

WhatNext Vision Motors is proposed as a customer-first mobility ecosystem connecting vehicle discovery, ownership, after-sales service, dealer operations and fleet intelligence.



Vision

- Make mobility simpler, safer, more connected and more sustainable.
- Provide a consistent experience across website, mobile app, dealer portal and fleet dashboard.

Business outcomes

- Improve lead conversion and test-drive attendance.
- Reduce service booking friction and missed maintenance.
- Increase operational visibility for dealers and fleet managers.
- Create a foundation for connected-vehicle and EV services.

Delivery approach

Start with a focused MVP, validate it with real users, then expand through measured releases.

2. Problem Statement & Opportunity

Customers often move between disconnected channels when researching vehicles, booking service or getting support. Dealers may use separate tools for leads, inventory and service work; fleet operators need timely operational insight.

Current pain points

- Scattered customer and vehicle information.
- Manual follow-ups and inconsistent status updates.
- Limited visibility into service capacity and parts readiness.
- Fleet maintenance triggered reactively rather than proactively.

Opportunity

- Create a shared digital foundation with role-based access and integration points.
- Use analytics to measure service quality, funnel conversion and utilization.

Success principle

Build only the capabilities that solve validated user problems; avoid collecting vehicle or personal data without a clear purpose.

3. Product Vision & Scope

The proposed product is a connected set of digital experiences, not a single screen. The initial scope focuses on workflows that can be delivered without depending on advanced autonomous-driving capabilities.

In scope for MVP

- Vehicle catalogue and enquiry capture.
- Customer profile and vehicle garage.
- Test-drive request and lead management.
- Service booking and reminders.
- Dealer operations portal.
- Basic fleet dashboard and reporting.

Later releases

- Connected diagnostics and predictive maintenance where supported.
- Charging discovery and EV journey tools.
- Secure over-the-air update orchestration if vehicle architecture and compliance permit.
- Mobility subscriptions and partner services.

Out of scope initially

- Autonomous-driving control.
- Safety-critical remote vehicle commands.
- Unverified real-time vehicle telemetry.

4. Stakeholders & User Roles

Clear ownership and access boundaries are essential. Each role should see only the information needed to perform its tasks.

Primary users

- Prospective buyer: researches vehicles and requests a test drive.
- Vehicle owner: manages vehicles, service and support.
- Dealer sales advisor: follows leads and schedules test drives.
- Service advisor: manages appointments, estimates and work orders.
- Fleet manager: monitors utilization, alerts and maintenance.
- Platform administrator: manages roles, configuration and audit records.

Governance

- Name a product owner and business sponsor.
- Assign data owners for customer, vehicle, dealer and service records.
- Agree escalation paths for security incidents and service outages.

Role	Main permissions	Key restriction
Customer	Own profile, vehicles, bookings	Cannot view another customer's records
Dealer staff	Assigned leads and service jobs	Dealer/region scope enforced
Fleet manager	Authorized fleet and reports	No unrelated customer data
Administrator	Configuration and audit	Privileged actions logged

5. Use Case Catalogue

The first release should prioritize frequent, measurable journeys. These use cases are proposed examples, not claims about existing WhatNext systems.

Priority use cases

- UC-01 Discover a vehicle and submit an enquiry.
- UC-02 Schedule a test drive and receive reminders.
- UC-03 Create a customer account and link a vehicle.
- UC-04 Book a service appointment.
- UC-05 Dealer manages leads and follow-ups.
- UC-06 Service advisor tracks work orders and parts.
- UC-07 Fleet manager monitors vehicle utilization.
- UC-08 Admin manages access, consent and audit logs.

Prioritization method

Score each use case by customer value, business impact, delivery effort, risk and dependency. Deliver high-value, lower-dependency workflows first.

ID	Actor	Outcome
UC-01	Prospective buyer	Enquiry captured with consent
UC-02	Buyer / dealer	Confirmed test-drive slot
UC-03	Vehicle owner	Vehicle linked to verified account
UC-04	Vehicle owner	Service booking confirmed
UC-05	Sales advisor	Next action and lead status visible
UC-07	Fleet manager	Exceptions and utilization visible

6. UC-01: Vehicle Discovery & Enquiry

A buyer browses available models, compares key specifications, selects a preferred dealer and submits contact details with a clear consent notice.

Preconditions

- Vehicle catalogue and dealer data are published.
- Required privacy notice and consent wording are approved.

Main flow

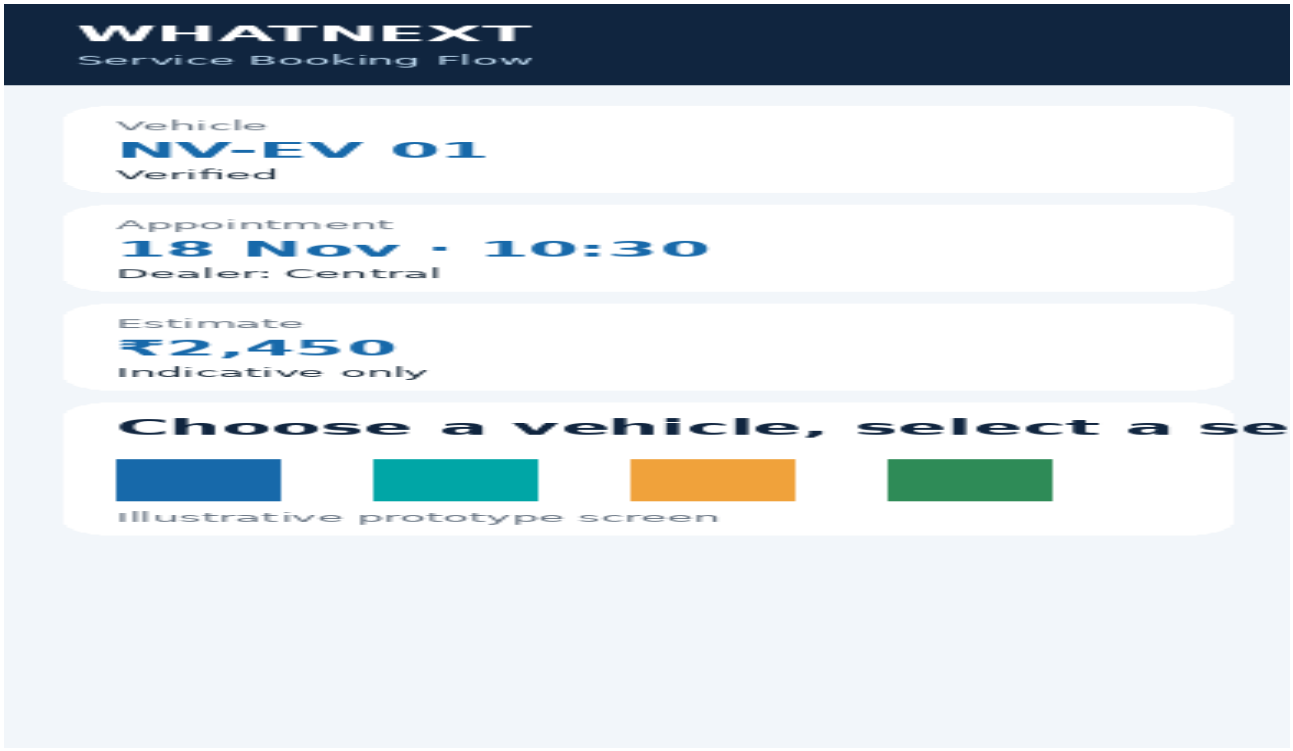
- Search or filter vehicles by model, body type, powertrain and budget.
- Open a vehicle detail page and compare specifications.
- Choose dealer and preferred contact method.
- Submit enquiry; system creates a lead ID and confirmation.
- Dealer receives the lead in the assigned queue.

Acceptance criteria

- Mandatory fields are validated.
- Duplicate submissions are detected or linked.
- Customer sees a confirmation reference.
- Lead creation and consent timestamp are auditable.

7. UC-02: Test-Drive Scheduling

The customer selects a model, dealer, date and available slot. The system confirms the request and provides rescheduling or cancellation instructions.



Main flow

- Check dealer availability and eligible vehicle demonstrator.
- Display only valid slots returned by the scheduling service.
- Capture driver/contact details required by policy.
- Confirm booking and notify customer and dealer.
- Send a reminder before the appointment.

Exceptions

- Slot becomes unavailable during booking: offer alternatives.
- Notification fails: retain booking and show status.
- Customer cancels: release slot and record the event.

Key metrics

- Booking completion rate.
- No-show rate.
- Time from enquiry to test drive.

8. UC-03: Customer Garage

The garage gives an owner a single view of linked vehicles, ownership documents where supported, service history, warranty dates and preferences.



Functional requirements

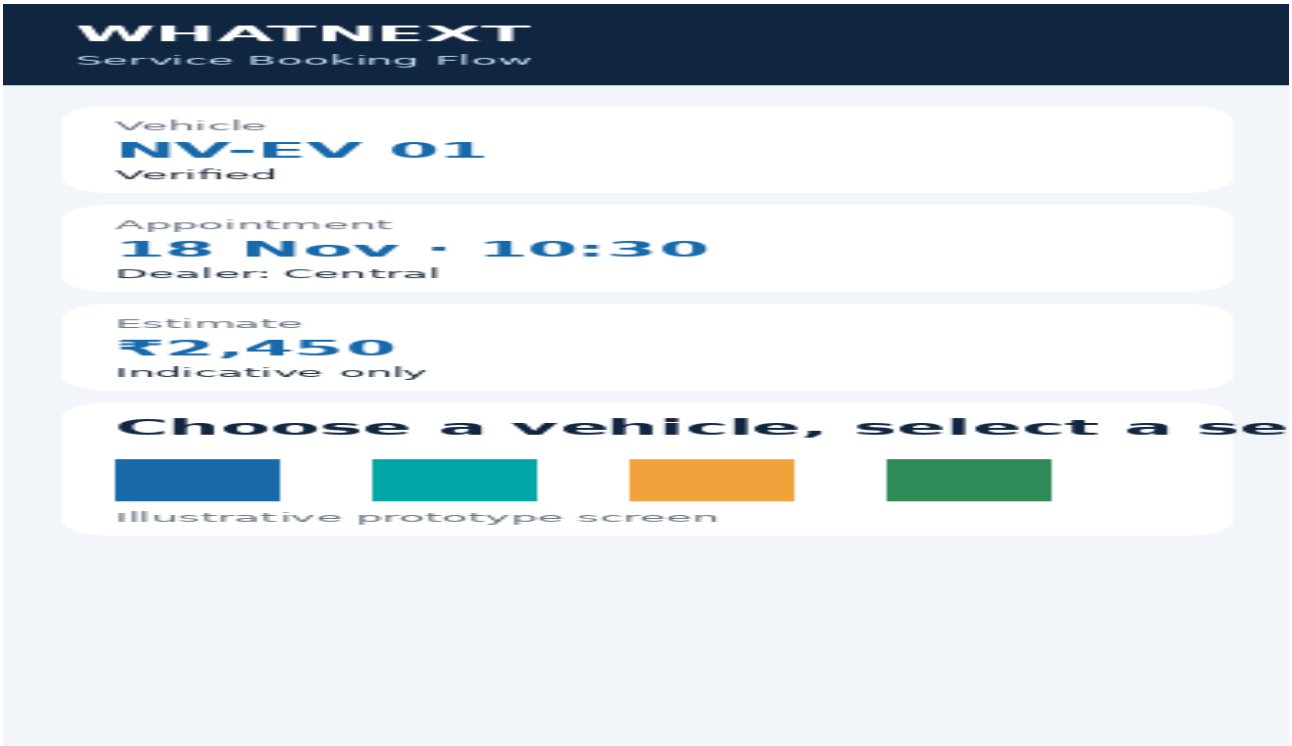
- Secure sign-in and account recovery.
- Vehicle linking through a verified ownership process.
- Show only permitted vehicle data.
- Display service timeline and next service reminder.
- Allow users to update preferences and manage consent.

Important controls

- Never expose full VIN or documents unnecessarily.
- Require step-up verification for sensitive changes.
- Provide a clear unlink-vehicle process.

9. UC-04: Service Booking & Reminders

The customer chooses a vehicle, service type, dealer and appointment slot. Estimates must be labelled as indicative until confirmed by the service team.



Functional requirements

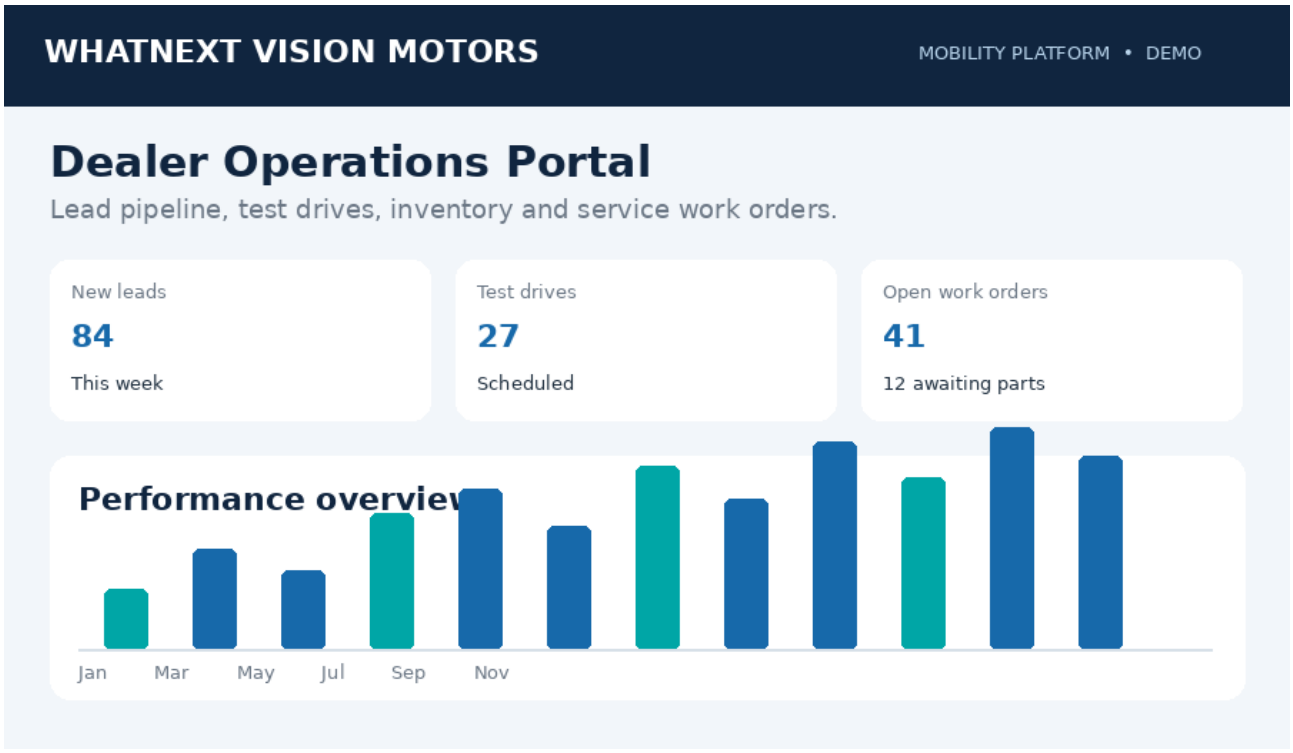
- Show eligible service types and required information.
- Display live slots only when connected to a trusted booking source.
- Provide confirmation, cancellation and rescheduling.
- Record booking status and customer communication preference.
- Support reminders based on time, mileage or verified diagnostics where available.

Business rules

- Do not imply a diagnostic finding without validated data.
- Explain estimate exclusions and approval steps.
- Respect consent and notification opt-outs.

10. UC-05: Dealer Lead Management

Dealer staff use a prioritized queue to manage enquiries, contact attempts, test drives and outcomes.



Lead statuses

- New → Assigned → Contacted → Qualified → Test drive → Won / Lost.
- Each transition records actor, timestamp and optional reason.
- Overdue follow-ups are highlighted.

Functional requirements

- Assign leads by dealer, territory and availability.
- Search/filter by status, model, source and date.
- Record notes and next-action date.
- Report response time and conversion funnel.

Controls

- Dealer users can access only assigned/authorized leads.
- Sensitive notes should be limited and governed.

11. UC-06: Service Work Orders & Parts

Service teams need a clear status flow from check-in through diagnosis, customer approval, repair, quality check and vehicle handover.

Workflow

- Appointment booked → Vehicle checked in → Diagnosis → Estimate approval → Repair → Quality check → Ready → Closed.

Requirements

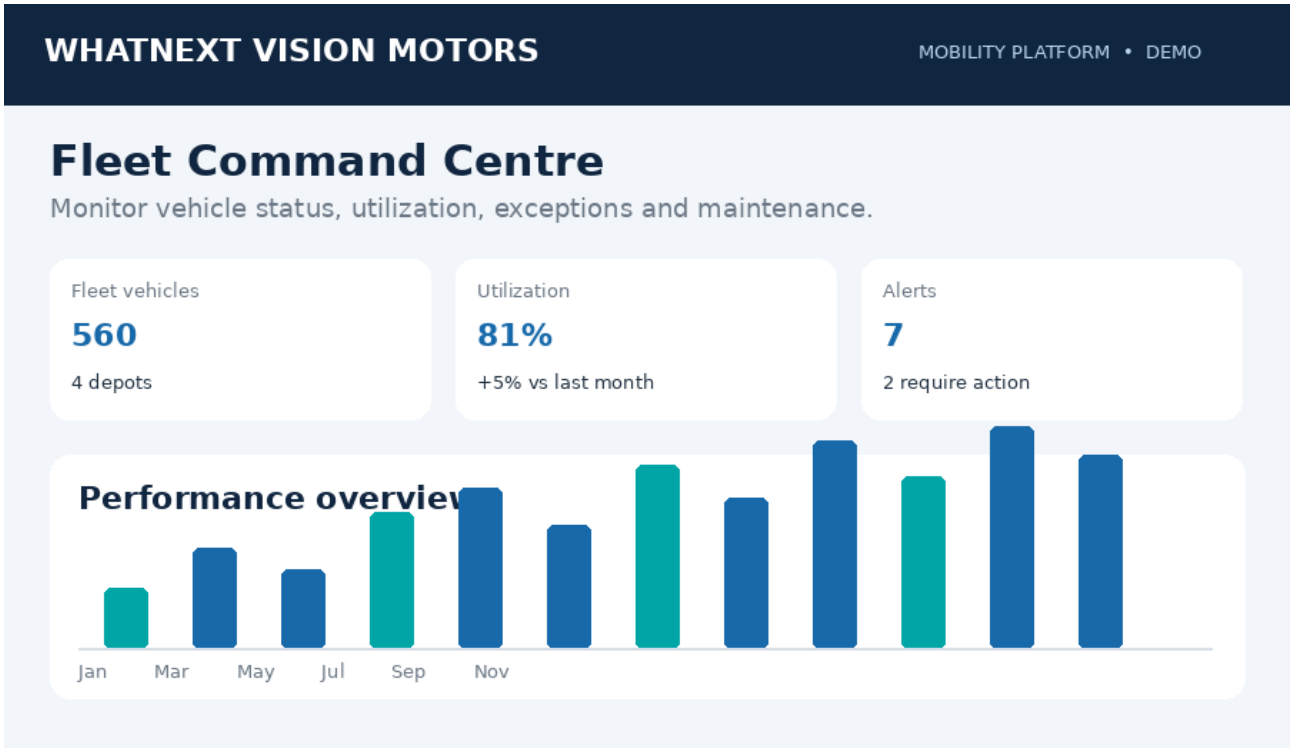
- Capture complaint and inspection checklist.
- Attach parts requests and expected availability.
- Notify customer when approval or pickup is required.
- Record labor, parts and final invoice references.
- Track SLA breaches and rework.

Risk control

Safety-related repairs require qualified review and documented sign-off.

12. UC-07: Fleet Operations

Fleet managers monitor vehicle availability, utilization, scheduled maintenance and operational exceptions. Connected data is optional and depends on device, network and contract support.



Dashboard capabilities

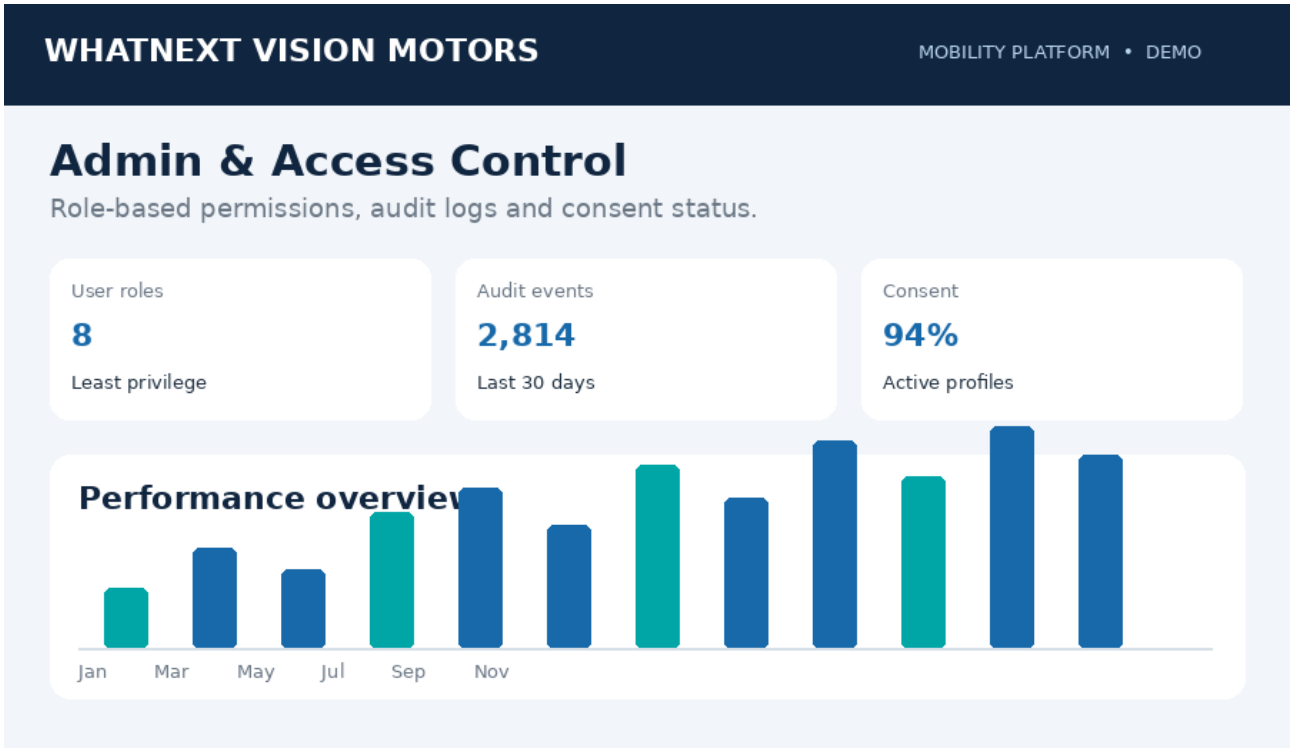
- Fleet inventory by depot and vehicle status.
- Utilization and downtime trends.
- Maintenance due list and service history.
- Exception queue for stale telemetry or fault alerts.
- Exportable reports for authorized users.

Data quality rules

- Display the last update timestamp.
- Mark stale or missing data clearly.
- Separate measured values from estimates.
- Avoid implying that an alert diagnoses a mechanical fault.

13. UC-08: Admin, Consent & Audit

Platform administrators maintain users, roles, dealer configuration, retention settings and audit visibility.



Requirements

- Role-based access control and least privilege.
- Multi-factor authentication for privileged roles.
- Consent records with purpose, version, timestamp and withdrawal.
- Audit log for access and sensitive changes.
- Configurable retention and deletion workflows.

Acceptance criteria

- Access denied by default when role is missing.
- Audit events include actor, action, target and time.
- Consent withdrawal propagates to applicable downstream uses.

14. Functional Requirements

Requirements should be uniquely identified, testable and linked to a use case and release.

MVP requirements

- FR-01: Browse and filter the vehicle catalogue.
- FR-02: Capture and track enquiries.
- FR-03: Request test-drive slots.
- FR-04: Authenticate users and manage profiles.
- FR-05: Link a vehicle to a verified owner.
- FR-06: Create, reschedule and cancel service bookings.
- FR-07: Manage dealer leads and follow-ups.
- FR-08: View service status and parts dependencies.
- FR-09: View fleet status and utilization.
- FR-10: Manage roles, consent and audit records.

Traceability

Each requirement should map to one or more user stories, test cases, release criteria and an accountable owner.

Req ID	Requirement	Priority
FR-01	Vehicle catalogue and filters	Must
FR-02	Lead capture and tracking	Must
FR-03	Test-drive scheduling	Should
FR-06	Service booking	Must
FR-09	Fleet dashboard	Should
FR-10	Access, consent and audit	Must

15. Non-Functional Requirements

Quality requirements protect customer trust and ensure the platform remains usable as adoption grows.

Performance & availability

- Set measurable response-time targets for key journeys.
- Define service availability targets and planned maintenance windows.
- Use caching carefully; do not cache sensitive data across users.

Security & privacy

- Encrypt traffic and sensitive stored data.
- Use secure secrets management, dependency scanning and security testing.
- Minimize personal data and define retention/deletion rules.

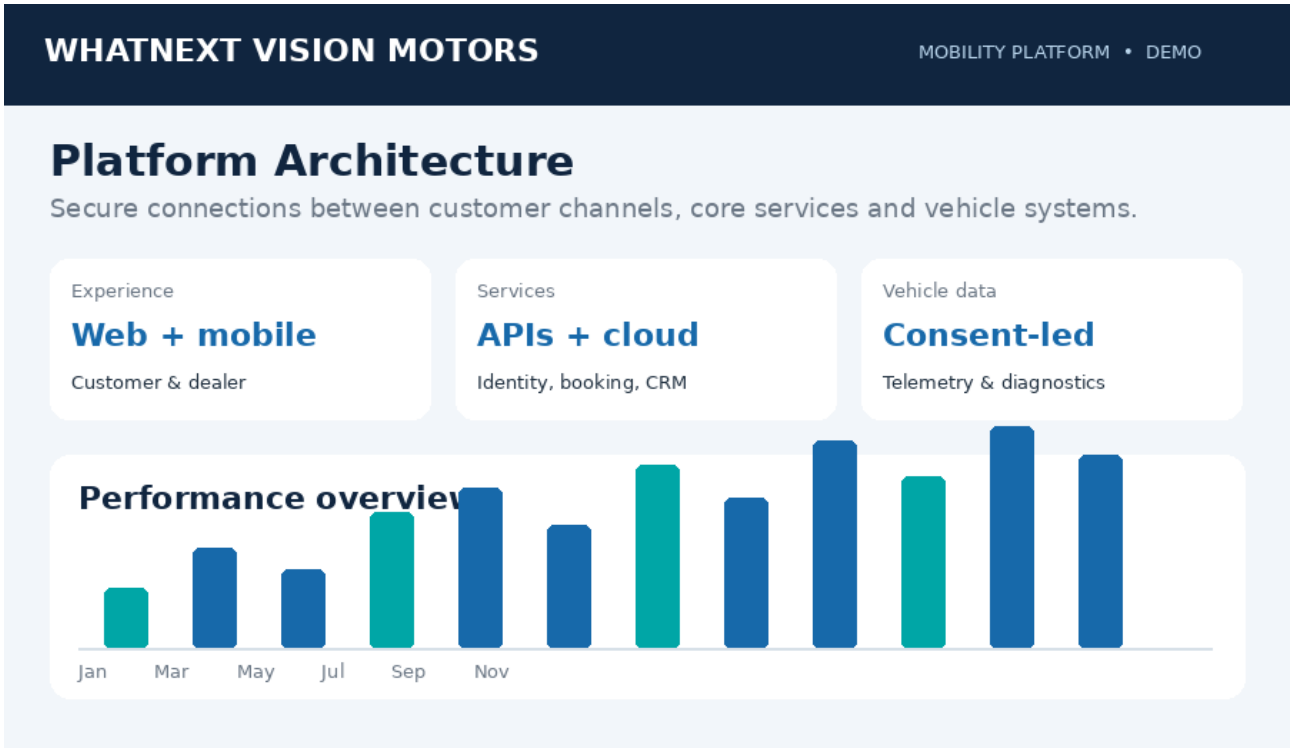
Accessibility & resilience

- Support keyboard navigation, readable contrast and screen-reader labels.
- Test on low-bandwidth mobile connections.
- Back up critical records and rehearse recovery.

Area	Proposed target for planning	Validation
Page performance	Core pages p95 under 2.5 s on agreed test profile	Synthetic and real-user tests
Availability	Agree target, e.g. 99.5% monthly for MVP	Monitoring report
Security	No unresolved critical/high findings at release	Independent test + remediation
Recovery	Agree RTO/RPO with business owner	Restore drill

16. Data Model & Integrations

Start with a small, well-governed domain model and integrate through documented APIs. Avoid tightly coupling the user interface to back-end systems.



Core entities

- Customer, Consent, Vehicle, Model, Dealer, Lead, TestDrive, Appointment, WorkOrder, PartRequest, Fleet, UserRole, AuditEvent.

Integration candidates

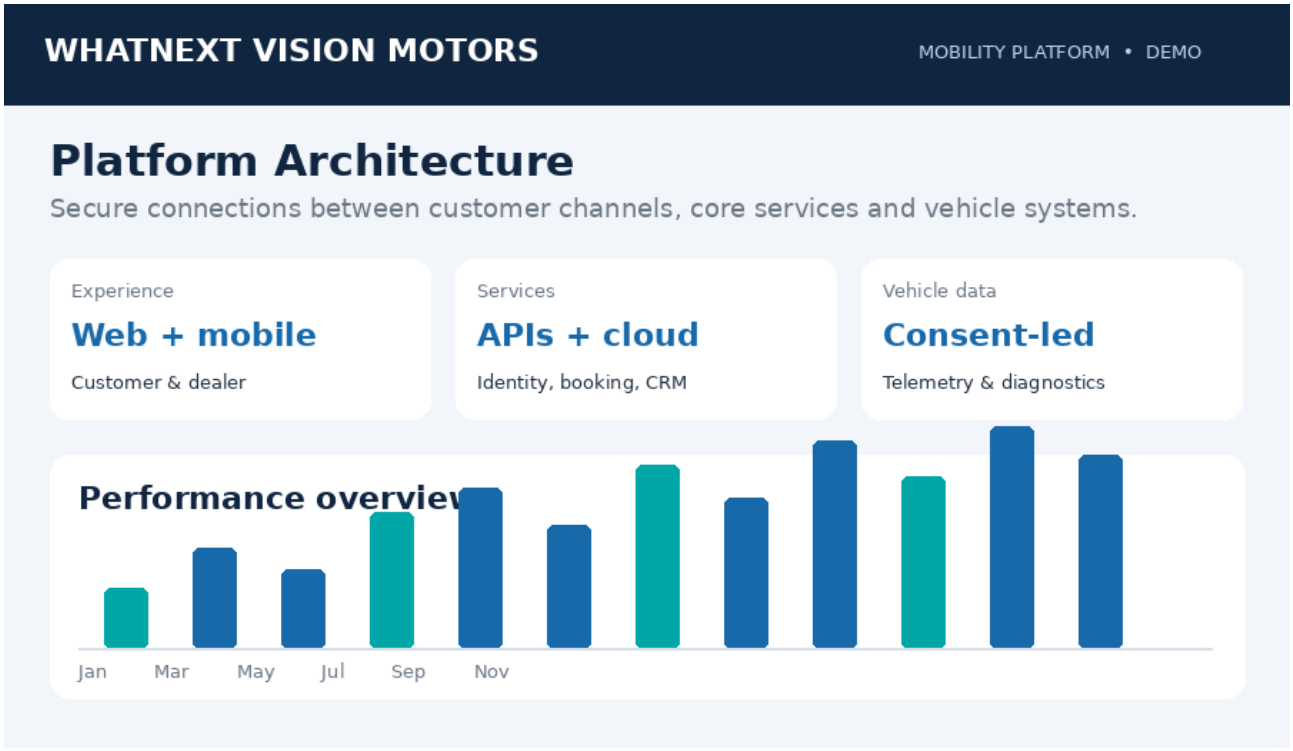
- CRM or lead management.
- Dealer management system (DMS).
- Service scheduling and parts inventory.
- Identity provider and notification service.
- Vehicle/telematics platform, only where authorized.
- Payment or finance partners in a later phase.

Integration approach

- API contracts with versioning and idempotency.
- Retries and dead-letter handling for failed events.
- Monitoring for data freshness and reconciliation.

17. Proposed Technical Architecture

Use a modular, API-first architecture so web, mobile and dealer experiences can share core services while keeping authorization centralized.



Layers

- Experience: responsive website, mobile app and dealer/fleet portal.
- API edge: authentication, rate limiting and request validation.
- Domain services: catalogue, leads, bookings, service, fleet, consent and notifications.
- Data layer: transactional store, analytics store and audit logs.
- Integration layer: CRM/DMS, dealer inventory and connected-vehicle systems.

Architecture principles

- Keep safety-critical vehicle control out of the MVP.
- Use event-driven integration where it improves reliability.
- Maintain environment separation for development, test and production.
- Design for observability and controlled feature rollout.

18. Security, Privacy & Compliance

Security and privacy are product requirements, not final-stage add-ons. Applicable obligations depend on markets, vehicle capabilities, data types and business model.

Baseline controls

- Threat modelling for customer, dealer, fleet and connected-vehicle flows.
- Strong identity, MFA for privileged access and role-scoped authorization.
- Encryption in transit and at rest; managed secrets and key rotation.
- Audit trails, incident response and vulnerability management.
- Consent, purpose limitation, retention and deletion workflows.

Automotive-specific considerations

- If connected-vehicle features are introduced, assess relevant automotive cybersecurity and software-update standards, such as ISO/SAE 21434 and ISO 24089, plus applicable regulations.
- Conduct legal and safety review before remote commands or OTA update functionality.

Reference

AWS Connected Mobility Lens discusses provisioning, OTA updates and automotive security design considerations; use it as a technical reference, not as a substitute for legal advice.

19. Scheduled Vehicle Order Processing

This section documents a working Salesforce implementation that automates recurring vehicle-order processing as a background job instead of a manual task. The screenshots are captured from the project's Salesforce Developer Edition org and show the real configuration, a scheduling error that was diagnosed and resolved, and the job registered in the Apex Jobs monitor.

Purpose and fit with the platform

- Runs vehicle-order processing automatically every day, supporting the dealer and operations workflows described in UC-05 and UC-06.
- Follows the API-first, event-driven principles of Section 17 and the data-freshness, retry and monitoring expectations of Section 16.
- Uses asynchronous Apex so processing runs outside user transactions, in controlled batches of 50 records.

Implementation components

Component	Type	Role
VehicleOrderTrigger	Apex trigger (.apxt)	Trigger entry point for vehicle order records
VehicleOrderTriggerhandler	Apex class (.apxc)	Handler class that holds the trigger logic
VehicleOrderBatch	Batch Apex class (.apxc)	Processes vehicle orders in batches
VehicleOrderBatchScheduler	Schedulable Apex class (.apxc)	Starts VehicleOrderBatch with Database.executeBatch(batchJob, 50)
Daily Vehicle Order Processing	Scheduled job name	Runs daily at 12:00 PM

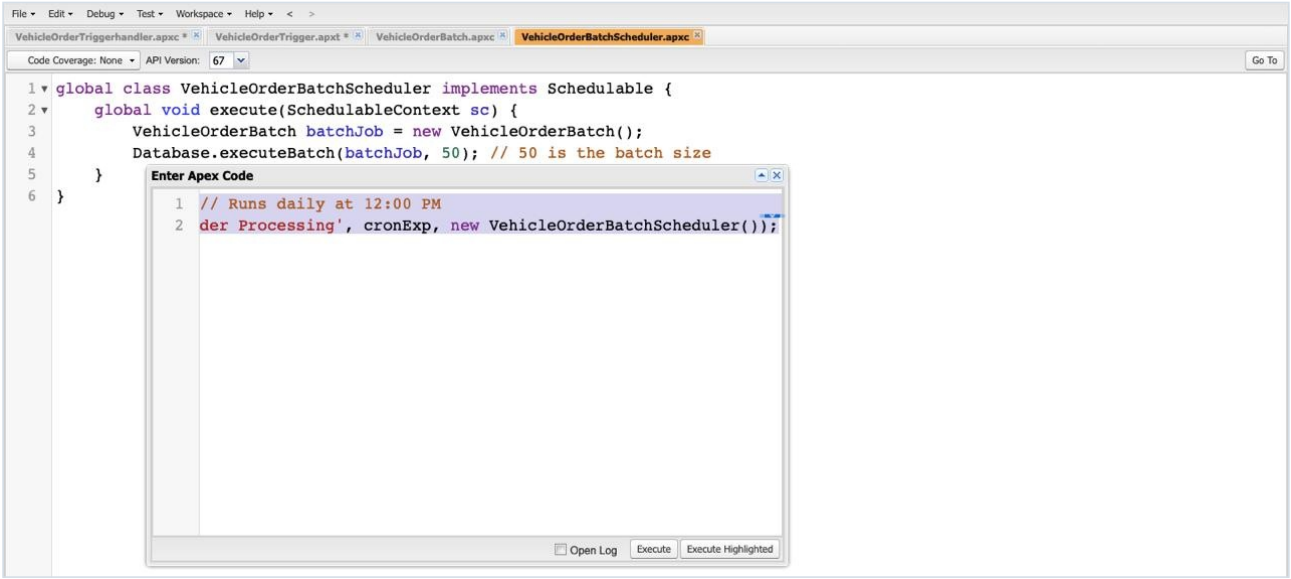


Figure 19.1 – VehicleOrderBatchScheduler class (batch size 50) and the Execute Anonymous window used to schedule it daily at 12:00 PM. The second line of the anonymous code is horizontally scrolled in the editor.

19.1 Scheduling Error & Resolution

When the schedule command was executed a second time, Salesforce rejected it because a job with the same name already existed. The error therefore confirms that the original scheduling had already succeeded.

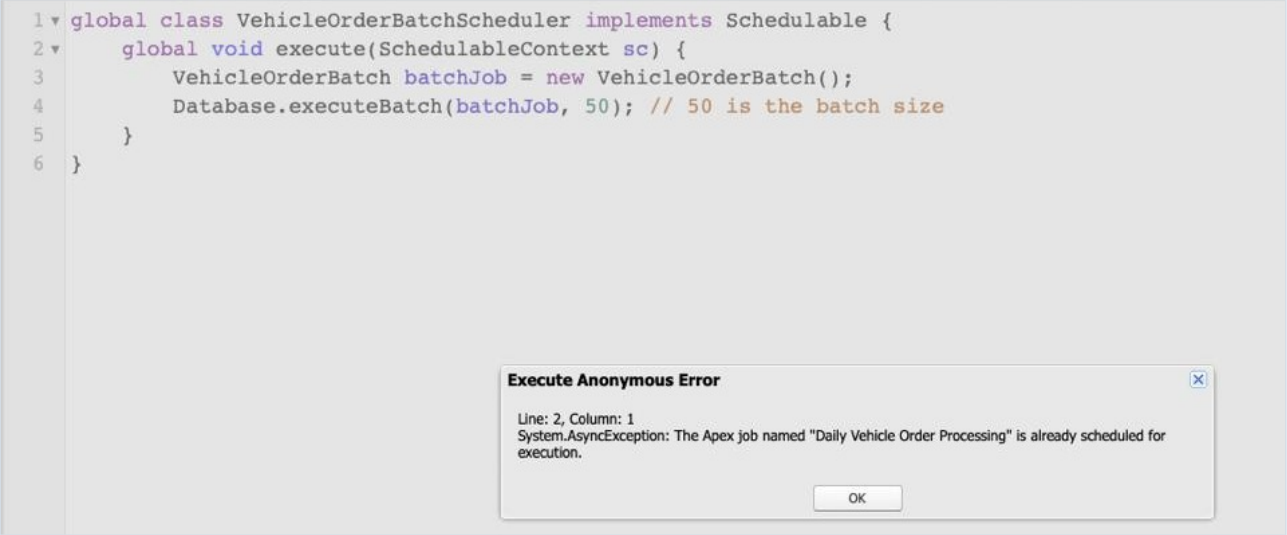


Figure 19.2 – Execute Anonymous Error: the Apex job named “Daily Vehicle Order Processing” is already scheduled for execution (Line 2, Column 1).

What the error means

- Exception: System.AsyncException, raised at Line 2, Column 1 of the anonymous code, which is the line that calls System.schedule.
- Scheduled job names must be unique, and “Daily Vehicle Order Processing” is already used by an existing scheduled job.
- The duplicate request was rejected, so no second schedule was created.

Resolution options

- Keep the existing job if its schedule is correct. Section 19.2 confirms it is registered, so no action is needed.
- To change the time or class version, delete the existing job under Setup → Scheduled Jobs, then run the schedule command again.
- Or schedule under a different, unique job name.
- For repeatable deployments, check for an existing job with that name (CronJobDetail) before calling System.schedule, or abort it first with System.abortJob.

19.2 Verification in Apex Jobs

The Apex Jobs page in Setup confirms that the schedule is registered and waiting for its next run.

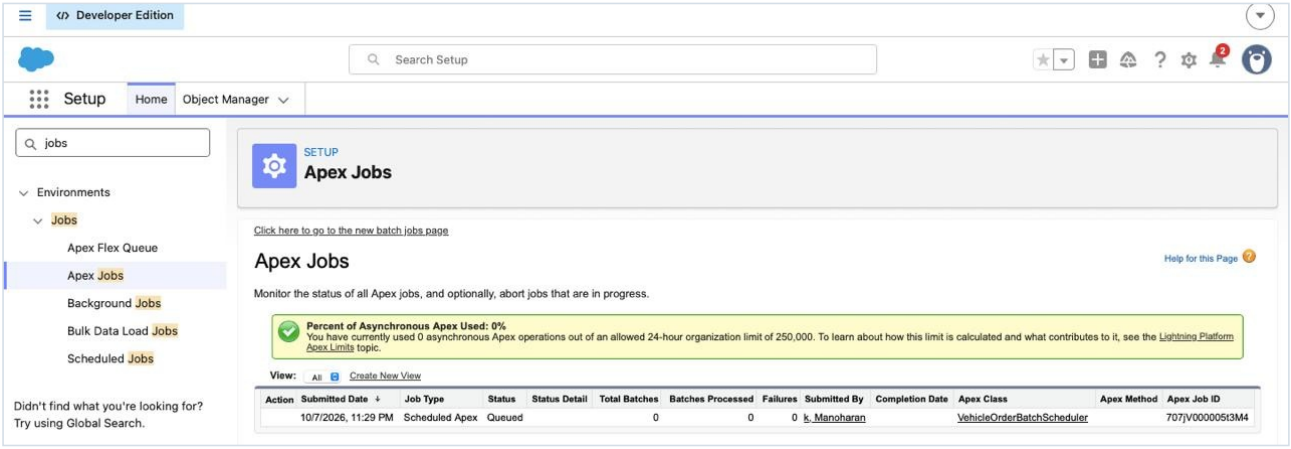


Figure 19.3 – Setup → Apex Jobs showing the Scheduled Apex job for VehicleOrderBatchScheduler in Queued status.

What the monitor shows

Field	Value
Job type / status	Scheduled Apex / Queued
Apex class	VehicleOrderBatchScheduler
Submitted	10/7/2026, 11:29 PM
Apex job ID	707jV000005t3M4
Total batches / processed / failures	0 / 0 / 0 (batches appear after the first 12:00 PM run)
Asynchronous Apex used	0% of the 250,000 operations allowed in 24 hours

Verification checklist

- The scheduled job exists exactly once under the name “Daily Vehicle Order Processing”.
- After the next 12:00 PM run, a Batch Apex row for VehicleOrderBatch appears with Total Batches above 0 and Failures at 0.
- Any failed batch is reviewed the same day, in line with the monitoring and runbook expectations of Sections 28 and 29.

Before production release

- Add Apex test classes for the trigger, handler, batch and scheduler. The Developer Console shows Code Coverage: None, and Salesforce requires at least 75% coverage to deploy to production.
- Save the two files marked with an asterisk (unsaved changes) in the Developer Console tabs: VehicleOrderTriggerhandler.apxc and VehicleOrderTrigger.apxt.
- Record the schedule and job-name convention in the operations runbook.

20. Screenshot Gallery: Customer App

This concept screen illustrates a vehicle garage with vehicle status, service information and an ownership summary. It is a designed mockup for requirements discussion, not a screenshot of a live application.

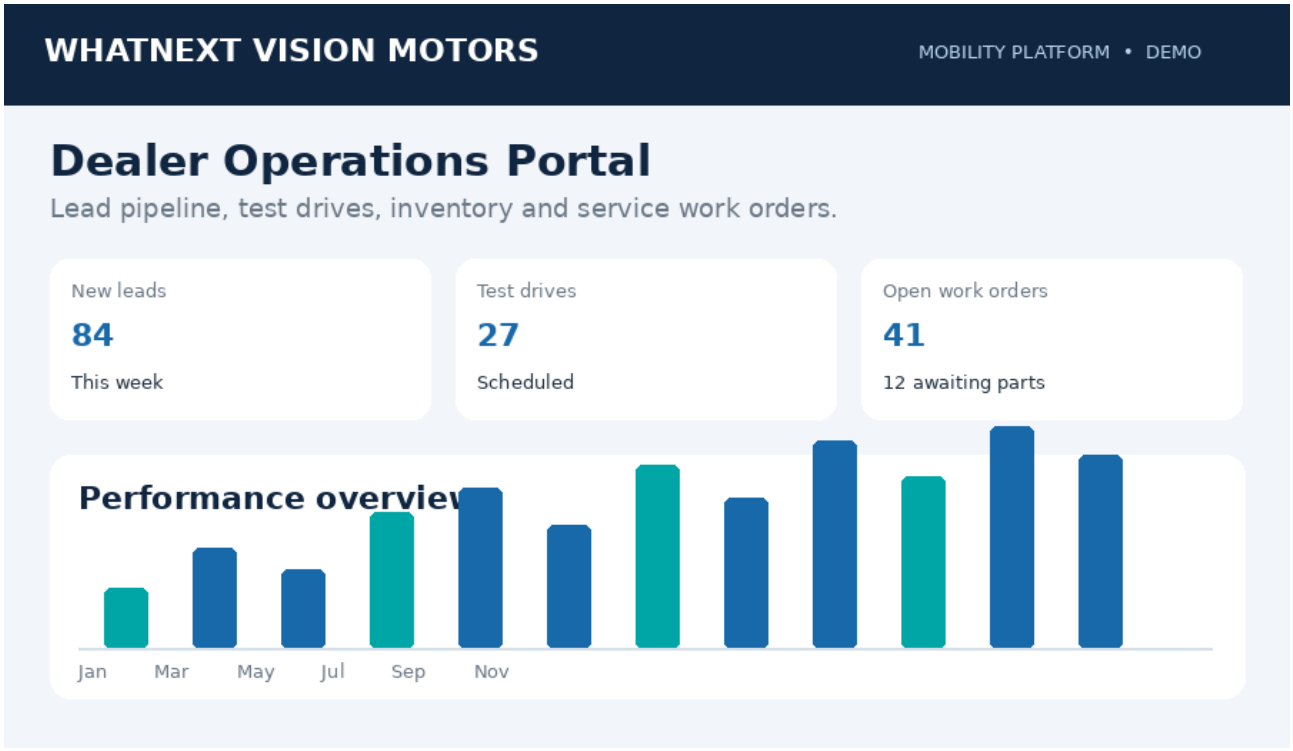


What to validate

- Is the most important vehicle status visible at a glance?
- Can a user find the next service action quickly?
- Are estimates and connected status explained clearly?

21. Screenshot Gallery: Dealer Portal

The dealer portal concept focuses on a work queue: leads, test drives and service jobs that require attention. Production versions should use real authorized data and consistent role permissions.

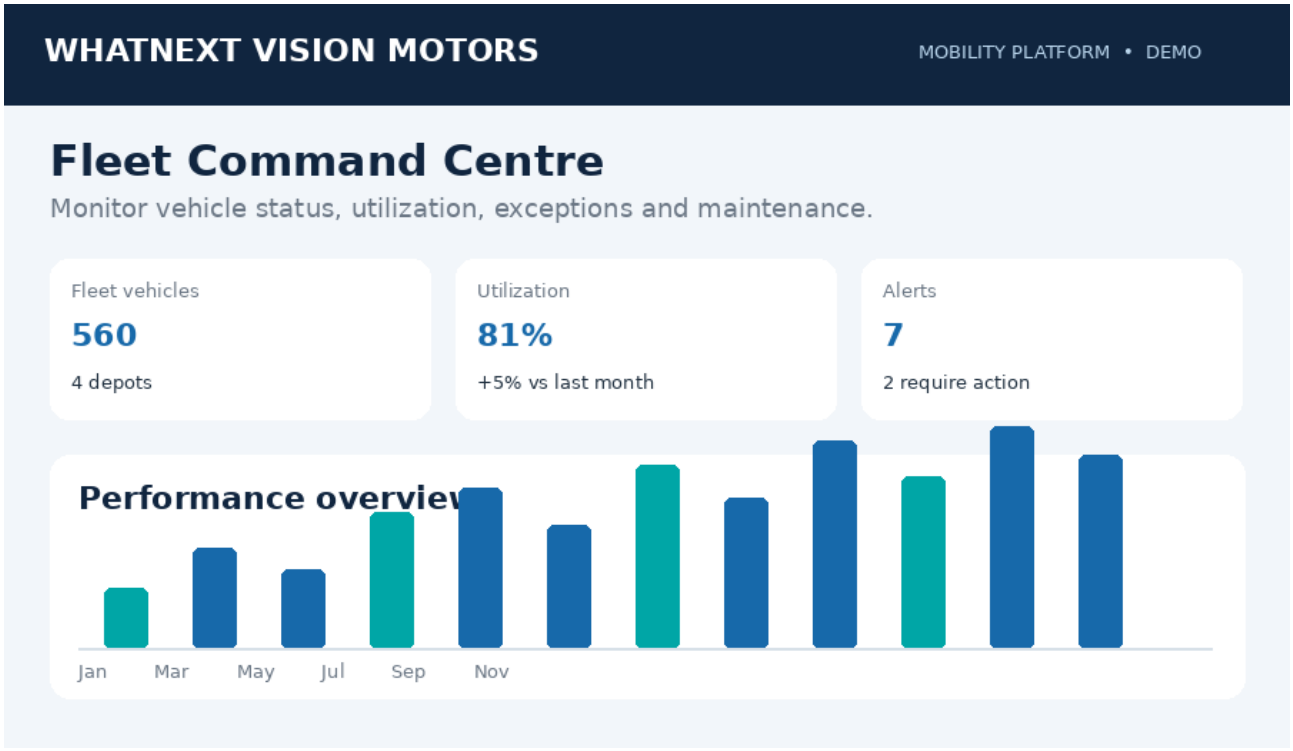


What to validate

- Can staff identify overdue actions quickly?
- Are lead ownership and next steps clear?
- Can the interface be used comfortably on dealership tablets and desktops?

22. Screenshot Gallery: Fleet & Sustainability

Fleet dashboards should distinguish operational measures from estimates and should show data freshness. Sustainability measures require a documented calculation method.

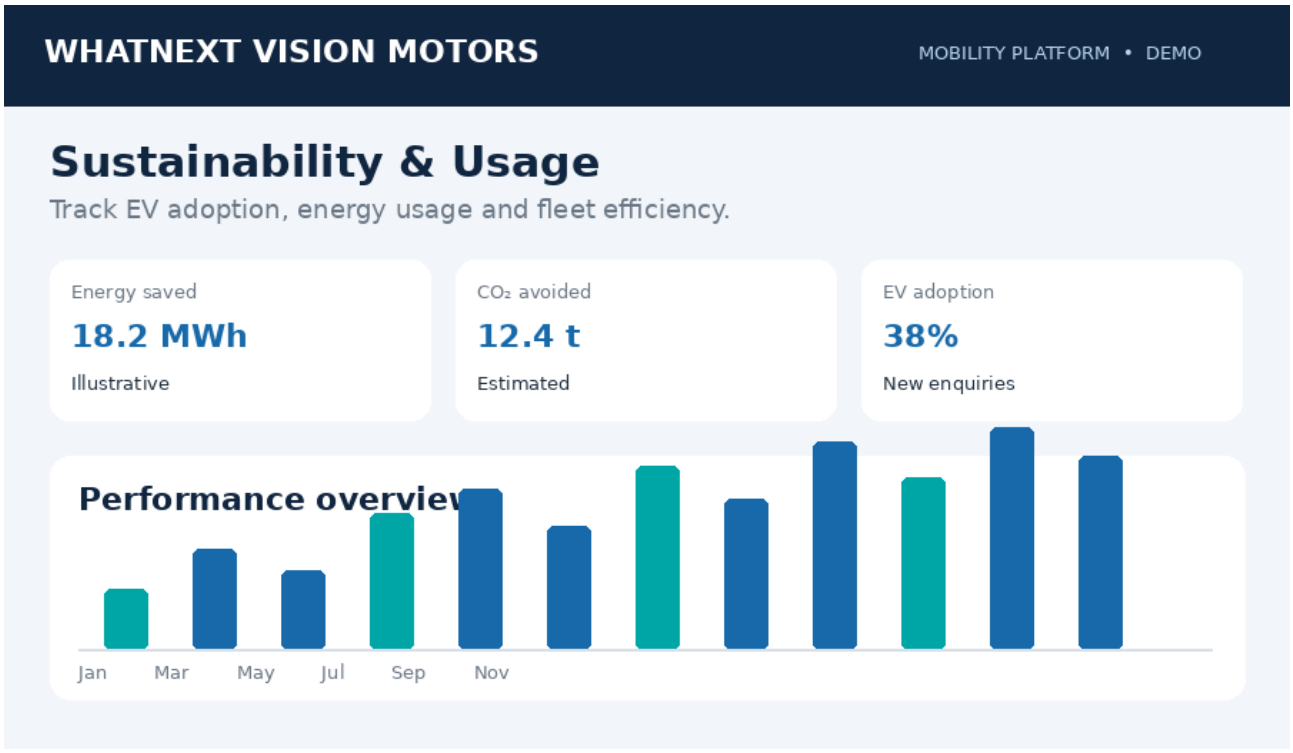


What to validate

- Are utilization and downtime definitions agreed?
- Can managers drill into an alert without exposing unrelated data?
- Are CO_2 and energy figures clearly marked as estimates when applicable?

23. Reporting & KPIs

Define KPI formulas before building dashboards so teams do not interpret the same metric differently.



Customer & sales

- Lead response time = first meaningful response minus lead creation time.
- Lead conversion = won leads divided by eligible leads in the cohort.
- Test-drive show rate = attended bookings divided by confirmed bookings.

Service & operations

- Booking completion and cancellation rates.
- On-time service completion and first-time fix rate.
- Fleet utilization and unplanned downtime.

Experience & trust

- Customer satisfaction/NPS using an agreed survey method.
- Consent withdrawal completion time.
- Security incident response and resolution time.

24. Delivery Method & Team

Use short delivery cycles with frequent stakeholder review. Keep product, engineering, design, QA, security and operations involved from discovery through launch.

Suggested core team

- Product owner / business analyst.
- UX/UI designer.
- Technical lead and 2–4 engineers, adjusted to scope.
- QA automation engineer.
- Cloud/platform engineer.
- Security/privacy reviewer and business SMEs.

Ways of working

- Two-week sprints with a demonstrable increment.
- Backlog refinement and acceptance criteria before development.
- Continuous integration, automated tests and peer review.
- Weekly risk/dependency review and fortnightly stakeholder demo.

25. Milestone Roadmap (24 Weeks)

A 24-week plan is a planning baseline, not a commitment. Actual dates depend on team size, integrations, procurement, compliance review and stakeholder availability.

Phase gates

- M1 — Discovery complete (Weeks 1–3): goals, personas, current-state workflows and success measures approved.
- M2 — Solution design (Weeks 4–6): UX prototypes, architecture, data model, threat model and backlog approved.
- M3 — MVP build (Weeks 7–14): catalogue, leads, identity, booking and dealer workflow implemented.
- M4 — Integration & hardening (Weeks 15–18): integration tests, accessibility, security, performance and data migration rehearsed.
- M5 — Pilot (Weeks 19–21): selected dealers and users; support and feedback loop active.
- M6 — Launch decision (Weeks 22–24): exit criteria met, training completed and go/no-go approved.

Milestone	Evidence	Owner
M1 Discovery	Signed problem statement + KPI baseline	Product owner
M2 Design	Approved prototype + architecture	Design / Tech lead
M3 Build	Demo + automated test report	Engineering
M4 Hardening	Security, performance, integration results	QA / Security
M5 Pilot	Pilot metrics + issue triage	Business lead
M6 Launch	Go/no-go checklist + support readiness	Sponsor

26. Sprint-Level Plan & Dependencies

The roadmap should be translated into a backlog with explicit dependencies and release criteria. Integration access is a frequent schedule driver.

Example sprint sequence

- Sprints 1–2: design system, identity, catalogue and test scaffolding.
- Sprints 3–4: lead capture, notifications and dealer queue.
- Sprints 5–6: test-drive and service booking flows.
- Sprints 7–8: work orders, admin, audit and reporting.
- Sprints 9–10: integrations, hardening, accessibility and pilot fixes.

Critical dependencies

- Availability of CRM/DMS APIs and test credentials.
- Dealer data quality and scheduling rules.
- Approved privacy notices, consent language and retention policy.
- Vehicle/telematics provider contracts if connected features are in scope.

Change control

New scope must include impact on time, cost, risk and MVP exit criteria.

27. Test Strategy & Release Readiness

Test continuously rather than leaving validation until the end. Test data must be synthetic or properly authorized and protected.

Test layers

- Unit tests for business rules and validation.
- API contract and integration tests.
- End-to-end tests for high-value customer journeys.
- Accessibility, mobile/responsive and browser coverage.
- Load, resilience, backup/restore and security testing.
- User acceptance testing with representative roles.

Release exit criteria

- No open critical defects; high defects assessed and accepted by accountable owners.
- All must-have acceptance criteria passed.
- Monitoring, alerts, support runbooks and rollback plan verified.
- Dealer and customer communications approved.

28. Risks, Assumptions & Mitigations

A risk register should be reviewed weekly and updated when assumptions change.

Key risks

- Integration APIs arrive late or behave inconsistently.
- Vehicle and customer data is incomplete or duplicated.
- Scope expands into connected-vehicle control too early.
- Dealer adoption is low due to workflow mismatch.
- Privacy/security requirements are discovered late.
- Metrics are launched without shared definitions.

Mitigations

- Run integration spikes early and maintain mocks for parallel development.
- Profile data and define ownership/deduplication rules.
- Keep safety-critical features out of MVP and require specialist review.
- Co-design workflows with pilot dealers and provide training.
- Perform threat modelling and privacy review during design.
- Publish a KPI dictionary and data-quality checks.

Risk	Likelihood / impact	Mitigation owner
API delay	Medium / High	Technical lead
Poor data quality	High / Medium	Data owner
Scope creep	Medium / High	Product owner
Low adoption	Medium / High	Dealer operations
Security/privacy gap	Low–Medium / High	Security/privacy lead

29. Operations, Support & Change Management

A successful launch includes support processes, staff readiness and a plan for iterative improvement.

Operational readiness

- Define support tiers, escalation contacts and severity levels.
- Monitor availability, latency, error rates and integration failures.
- Create runbooks for booking failures, identity issues and data reconciliation.
- Schedule backups and recovery exercises.
- Plan release notes, feature flags and rollback procedures.

Adoption plan

- Train pilot dealers using role-specific guides.
- Provide in-product help and short workflow videos.
- Collect feedback through structured surveys and support tickets.
- Review adoption and KPI trends weekly during pilot.

30. Recommendations & Next Steps

Proceed with a discovery-led MVP. Confirm business priorities and integration constraints before committing to a fixed budget or launch date.

Immediate next steps

- 1. Appoint executive sponsor and product owner.
- 2. Confirm target customer segments and MVP use cases.
- 3. Map existing CRM, dealer, service and fleet systems.
- 4. Approve data/privacy principles and security baseline.
- 5. Validate the mock screens with representative users.
- 6. Convert requirements into an estimate-ready backlog.
- 7. Agree milestone gates, budget range and pilot dealers.

Definition of MVP success

- Users can complete priority journeys end-to-end.
- Dealers consistently process leads and bookings in the new workflow.
- Security, privacy, performance and support exit criteria are met.
- Pilot KPIs are measured against a documented baseline.

References

- SAE International, “Next-Gen Automotive: Enablers and Monetization for Automotive SaaS and Software Defined Vehicles” (2026).
- Adobe Experience League, “Automotive use cases” (updated 2026).
- AWS, “Connected Mobility Lens” and vehicle/user provisioning guidance.
- Microsoft Learn, “Automotive connected fleets reference architecture.”

Decision needed	Options	Recommended starting point
Initial audience	Retail customers / dealers / fleets	Retail + dealer MVP; fleet dashboard as a limited module
Connected vehicle	Now / later	Later, after data access, consent and safety review
Pilot scale	1 dealer / 3–5 dealers / broad rollout	3–5 representative dealers if operationally feasible